

METHODS OVERVIEW

Procedure-agnostic post-selection inference for identified subgroups

A multiplier (wild / martingale) bootstrap recasting of the León (2024) de-biased estimator and infinitesimal-jackknife interval

forestsearch · Cox & GLM (DINA) estimands

THE PROBLEM

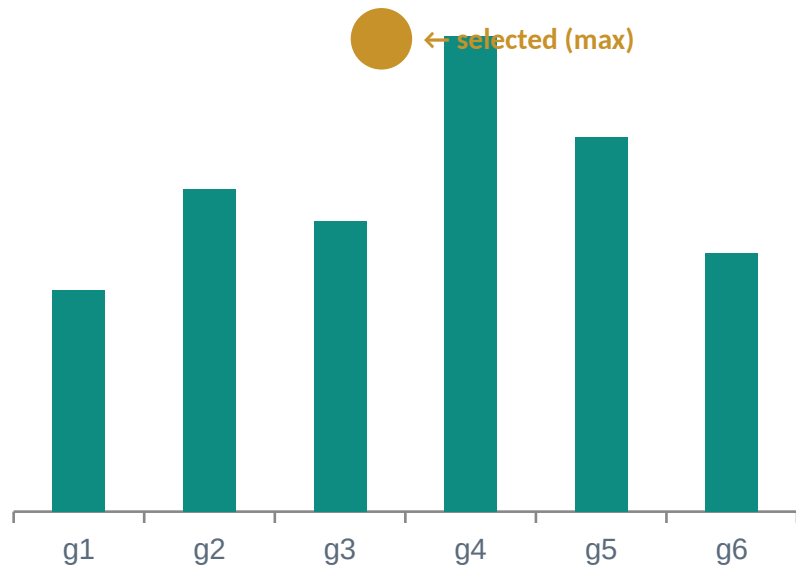
Selection optimism — the winner's curse

A subgroup chosen *because* its effect is extreme has an upward-biased effect. A naive within-subgroup CI, ignoring that the subgroup was selected, **under-covers**.

The inferential estimand

$\hat{\beta}(g)$ — the within-subgroup treatment effect on the natural-parameter scale (log-HR, log-OR, log-IRR, mean difference).

Candidate effects $\hat{\beta}(g)$ — the largest is selected



A two-term bootstrap correction, with an IJ interval

$$\tilde{\beta}(H) = \hat{\beta}(H) - (1/B) \sum_b [\eta^*_b(H^*_b) + \eta^*_b(H)]$$

$$\eta^*_b(g) = \hat{\beta}^*_b(g) - \hat{\beta}(g) \text{ (bootstrap minus observed effect at subgroup } g \text{)}$$

TERM 1

Selection optimism

Discrepancy at the re-selected subgroup — the winner's curse the bootstrap re-measures on each resample.

TERM 2

Estimation bias at H

Discrepancy at the fixed observed subgroup — the estimator's own bias there, and the IJ residual.

The bias term Harrell (1996) does not address

Harrell, Lee & Mark (1996)

Optimism correction

- **Term 1 only**
- Corrects selection optimism
- Estimator's own bias at H left uncorrected

León et al. (2024)

Two-term + IJ variance

- **Term 1 + Term 2**
- Adds $\eta^*(H)$: bias of $\hat{\beta}$ at the selected subgroup
- Makes $\tilde{\beta}$ a proper bagged estimator
- Supplies the residual for the IJ variance

A wild / martingale bootstrap of influence functions

1

One resample = one reweighting

Subject i gets multiplicity $K^*(b,i)$; all the randomness is one weight vector.

2

A refit = a dot product

$\hat{\beta}^*(g) - \hat{\beta}(g) = \Sigma (K^*(b,i) - 1) \cdot \text{dfbeta}(g,i) = D(g)$ — no refit.

3

One shared vector, all candidates

Overlapping subgroups move together → faithful selection competition.

4

Centered Poisson = the bootstrap weight

$K^*(b,i) - 1 \rightarrow \text{Poisson}(1) - 1$: the IJ linearization of the bootstrap.

→ The screen, the family routine, and the de-biasing all run from one $N \times S$ influence matrix.

The fast estimator IS León's Eq. 7, linearized

$$\eta^*_b(\mathbf{g}) = \sum_{i \in \mathbf{g}} (\mathbf{K}^*_{bi} - 1) \cdot \text{dfbeta}_{\mathbf{g},i} = \mathbf{D}_{\mathbf{g}}(\mathbf{b})$$

- **Point estimate:** $\tilde{\beta}(\mathbf{H})$ equals León's Eq. 7 to first order — for $\mathbf{H}$ and its complement $\mathbf{H}^c$.
- **Interval:** León's IJ variance from the same draws; wider than the naive SE (it carries selection variance).
- **Only approximation:** The dfbeta linearization — exact for mean-difference, loosest at small subgroup size.

WHAT "PROCEDURE-AGNOSTIC" MEANS

The inference wraps any selection on the effect

$$\hat{H} = \mathcal{I}(\{ \hat{\beta}(g) : g \in \mathcal{F} \})$$

TIER 1 • universal

Literal bootstrap re-run

Defined for any procedure you can re-run on a resample. The general definition of the de-biased estimator + IJ variance.

TIER 2 • fast & exact

Multiplier surrogate

First-order exact whenever the selection ranks on the inferential effect $\hat{\beta}(g)$ over an enumerable family.

THE LIMITATION

Model-based selectors must be re-ranked on the effect

GRF, DINA, and Lasso rank on their own statistic — not the Cox/GLM effect. They become candidate generators; selection moves to $\hat{\beta}(g)$.

GRF

native statistic

ranks regions by a mean doubly-robust (AIPW) score

DINA

native statistic

ranks by subgroup-mean per-patient effect $\hat{\tau} = \bar{a} \cdot \hat{\beta}_D$

Lasso

native statistic

selects a rule along a penalized coefficient path

Given up: the native ranking's own optimality as the selection criterion. **Gained:** an exactly correctable, validly covered effect.

WHY IT IS FORCED, NOT TUNING

The per-draw winner flip (worked example)

Two candidates — effect and native statistic disagree:

	g_1	g_2	selects
within-group effect $\hat{\beta}(g)$	0.90	1.00	effect $\rightarrow g_2$
DINA subgroup-mean $\hat{\tau}$	0.55	0.50	native $\rightarrow g_1$

Different winners on the same draw. The effect gate credits the wrong optimism for the native procedure — so its selection must rank on $\hat{\beta}(g)$.

On one resample b: $D_1=+0.05$, $D_2=+0.30$ (effect) • $E_1=+0.12$, $E_2=+0.02$ (native)

Native bootstrap (what DINA does)

T+E: g_1 0.67 > g_2 0.52 $\rightarrow$ picks g_1
credits $D_1 = +0.05$

Effect gate (re-selects on $\hat{\beta}+D$)

$\hat{\beta}+D$: g_1 0.95 < g_2 1.30 $\rightarrow$ picks g_2
credits $D_2 = +0.30$

Align the criterion — exact, conditional on the family

`select_statistic = "effect"` $\Rightarrow$ $T = \hat{\beta}$, $E = D$, one influence matrix, gate exact

What is proven

- DINA / GRF / Lasso $\rightarrow$ candidate generators
- Selection layer identical to forest search
- De-biased point + IJ interval first-order exact

The scope condition

- Conditional on the proposed family $\mathcal{F}$
- Exact for a fixed / enumerated family
- Data-driven $\mathcal{F}$ (DINA surface / GRF forest): family-regeneration component is the open piece

Validated on real data; one piece left to quantify

$\approx 637\times$

faster than the 500-boot bootstrap
(GBSG/Cox)

1.81

de-biased HR, gate — vs 1.93 full
bootstrap

0.48 / 0.36

IJ SE vs naive robust SE (log-HR)

Gate CI [0.71, 4.67] vs bootstrap [0.88, 4.24] — point estimates track; the IJ interval is mildly conservative at small n s.

Open (Q3): under a data-driven family, the fast gate omits the family-regeneration variance. Quantify it against a literal Tier-1 that re-runs DINA/GRF per resample — the León (2024) Table 2 design, extended with a Tier-2 block and a paired agreement panel.

CONTRIBUTIONS

One correction, recast and generalized

1

Procedure-agnostic

León's de-biasing + IJ wrap any effect-ranked selection.

2

Cox and GLM

One natural-parameter (DINA) scale; one influence representation.

3

Fast, exact to first order

Refit-free multiplier bootstrap = León's Eq. 7 (point + IJ).

4

A clean home for DINA/GRF

Aligned to the effect — exact conditional on the family.

Next: simulate operating characteristics (bias · coverage · calibration) and the family-regeneration gap.